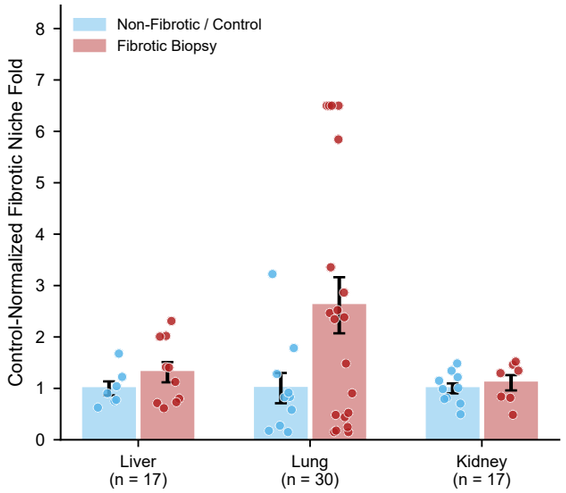
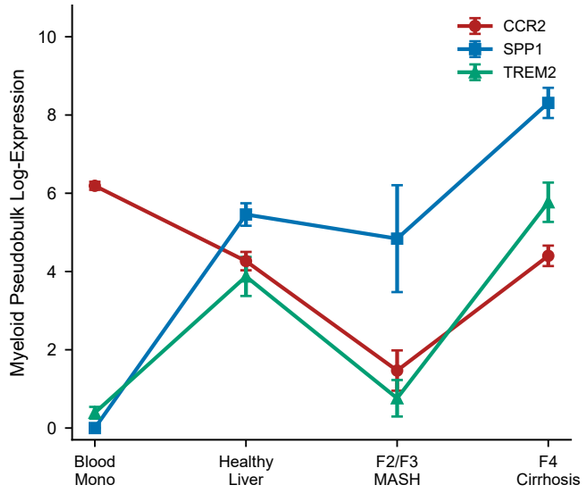
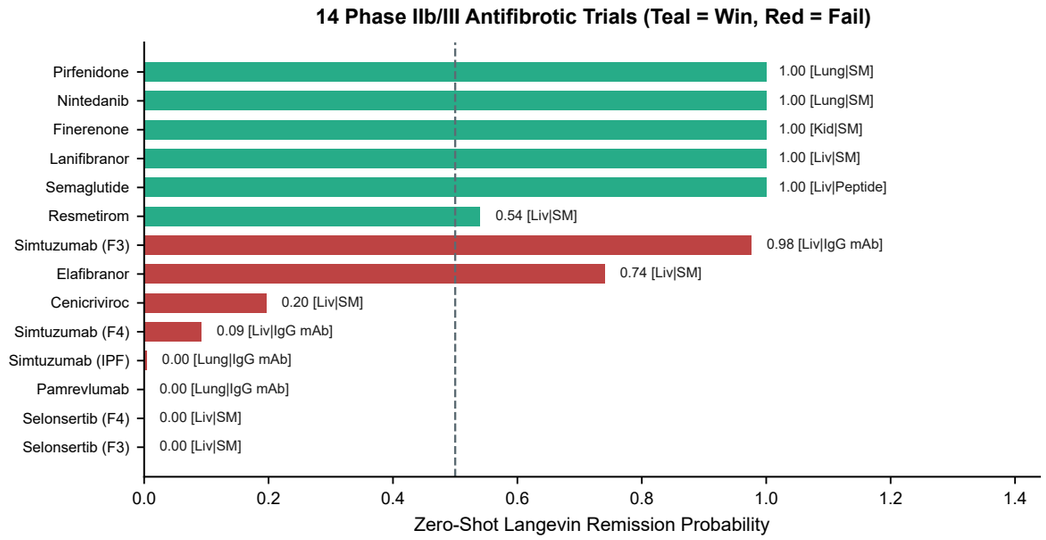
a


64-Donor 3-Organ Fibrosis Atlas
Non-Fibrotic / Control
Fibrotic Biopsy
Control-Normalized Fibrotic Niche Fold
Liver (n = 17)
Lung (n = 30)
Kidney (n = 17)

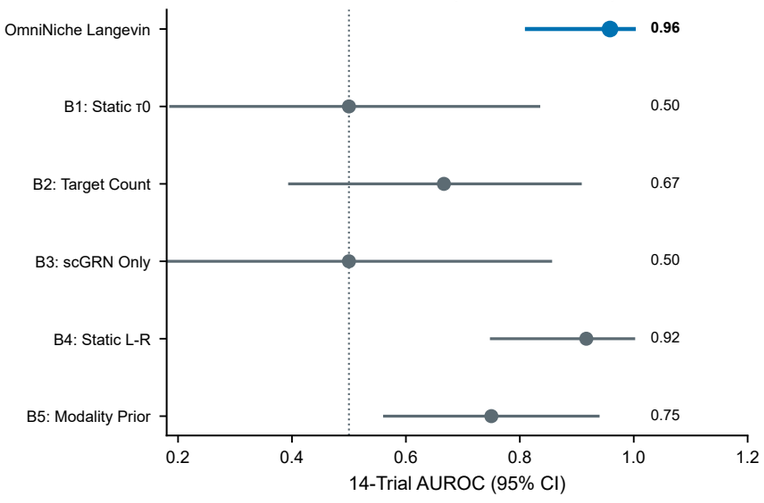
b

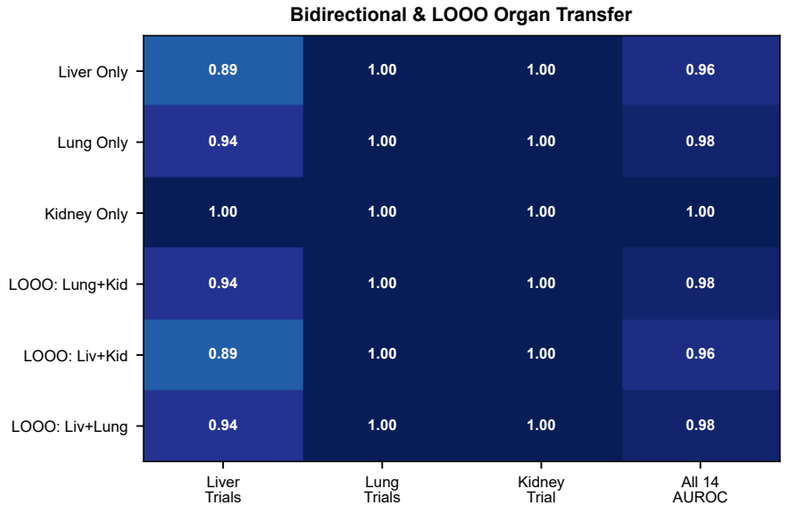

CCR2 Collapse vs. SAMac Emergence
CCR2
SPP1
TREM2
Myeloid Pseudobulk Log-Expression
Blood Mono
Healthy Liver
F2/F3 MASH
F4 Cirrhosis

c

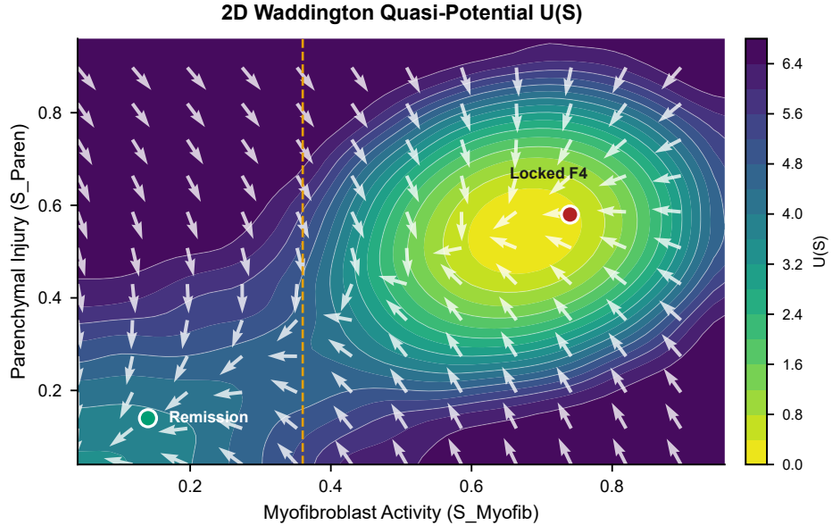

14 Phase IIb/III Antifibrotic Trials (Teal = Win, Red = Fail)
Pirfenidone 1.00 [Lung|SM]
Nintedanib 1.00 [Lung|SM]
Finerenone 1.00 [Kid|SM]
Lanifibranor 1.00 [Liv|SM]
Semaglutide 1.00 [Liv|Peptide]
Resmetirom 0.54 [Liv|SM]
Simtuzumab (F3) 0.98 [Liv|IgG mAb]
Elafibranor 0.74 [Liv|SM]
Cenicriviroc 0.20 [Liv|SM]
Simtuzumab (F4) 0.09 [Liv|IgG mAb]
Simtuzumab (IPF) 0.00 [Lung|IgG mAb]
Pamrevlumab 0.00 [Lung|IgG mAb]
Selonsertib (F4) 0.00 [Liv|SM]
Selonsertib (F3) 0.00 [Liv|SM]
Zero-Shot Langevin Remission Probability

d

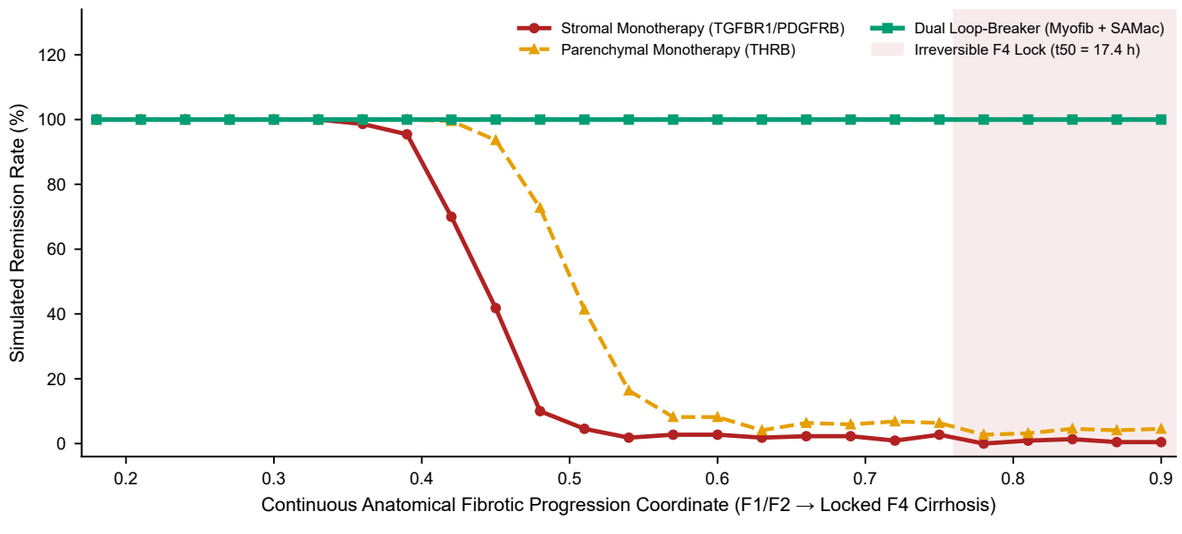

Zero-Shot Arena (Perm. P = 0.0028)
OmniNiche Langevin 0.96
B1: Static τ0 0.50
B2: Target Count 0.67
B3: scGRN Only 0.50
B4: Static L-R 0.92
B5: Modality Prior 0.75
14-Trial AUROC (95% CI)


Bidirectional & LOOO Organ Transfer
Liver Only 0.89 1.00 1.00 0.96
Lung Only 0.94 1.00 1.00 0.98
Kidney Only 1.00 1.00 1.00 1.00
LOOO: Lung+Kid 0.94 1.00 1.00 0.98
LOOO: Liv+Kid 0.89 1.00 1.00 0.96
LOOO: Liv+Lung 0.94 1.00 1.00 0.98
Liver Trials
Lung Trials
Kidney Trial
All 14 AUROC

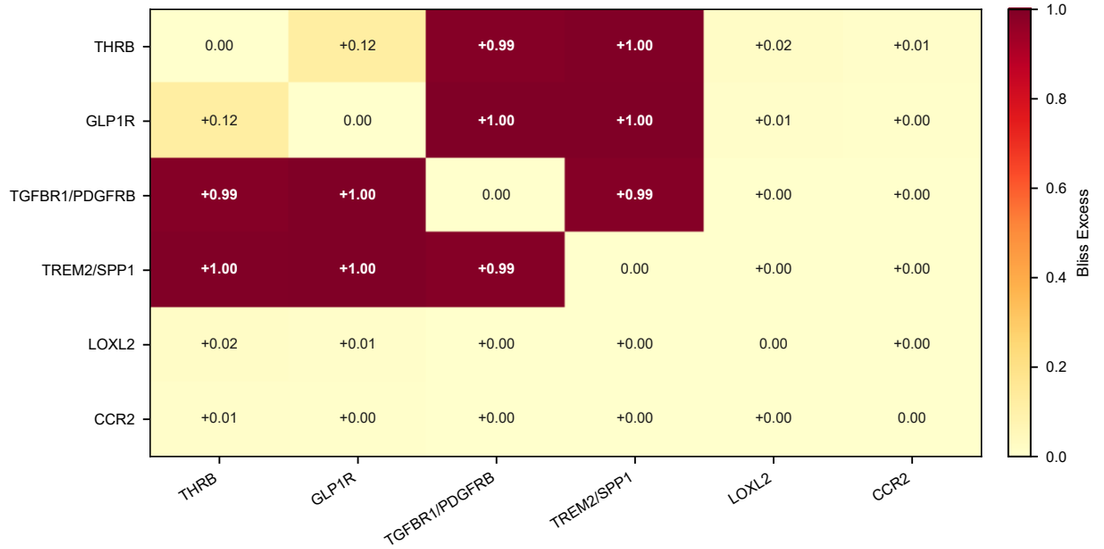
e


2D Waddington Quasi-Potential U(S)
Locked F4
Remission
Parenchymal Injury (S_Paren)
Myofibroblast Activity (S_Myofib)
U(S)

f


Hysteresis Collapse of Late-Stage Monotherapy & Dual Rescue
Stromal Monotherapy (TGFBR1/PDGFRB)
Parenchymal Monotherapy (THRB)
Dual Loop-Breaker (Myofib + SAMac)
Irreversible F4 Lock (t50 = 17.4 h)
Simulated Remission Rate (%)
Continuous Anatomical Fibrotic Progression Coordinate (F1/F2 → Locked F4 Cirrhosis)

g


Locked-F4 2D Bliss Synergy Excess
THRB 0.00 +0.12 +0.99 +1.00 +0.02 +0.01
GLP1R +0.12 0.00 +1.00 +1.00 +0.01 +0.00
TGFBR1/PDGFRB +0.99 +1.00 0.00 +0.99 +0.00 +0.00
TREM2/SPP1 +1.00 +1.00 +0.99 0.00 +0.00 +0.00
LOXL2 +0.02 +0.01 +0.00 +0.00 0.00 +0.00
CCR2 +0.01 +0.00 +0.00 +0.00 +0.00 0.00
Bliss Excess